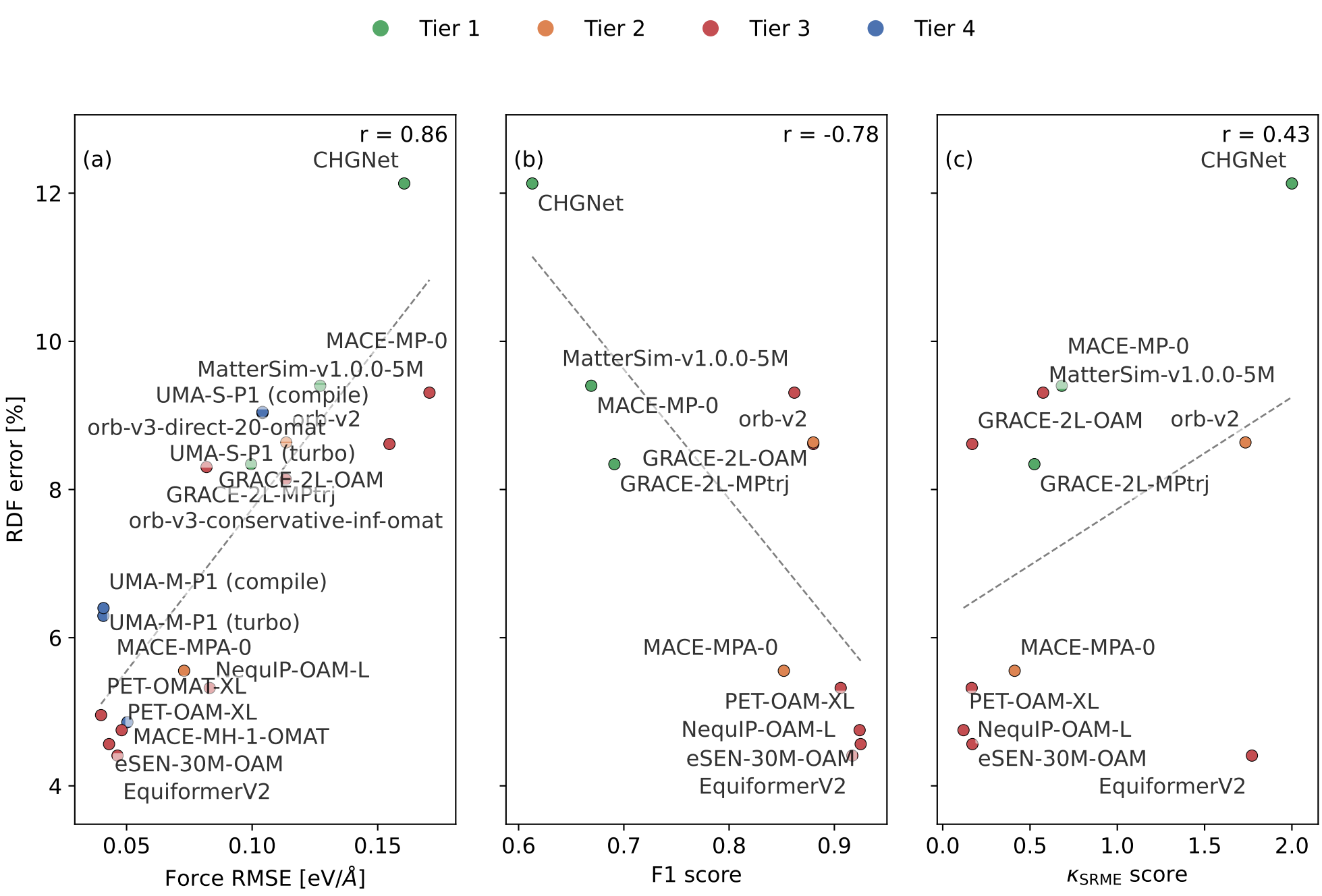
Tier 1
Tier 2
Tier 3
Tier 4
(a)
r = 0.86
RDF error [%]
Force RMSE [eV/Å]
CHGNet
MACE-MP-0
MatterSim-v1.0.0-5M
UMA-S-P1 (compile)
orb-v3-direct-20-omat
orb-v2
UMA-S-P1 (turbo)
GRACE-2L-OAM
GRACE-2L-MPtrj
orb-v3-conservative-inf-omat
UMA-M-P1 (compile)
UMA-M-P1 (turbo)
MACE-MPA-0
NequIP-OAM-L
PET-OMAT-XL
PET-OAM-XL
MACE-MH-1-OMAT
eSEN-30M-OAM
EquiformerV2
(b)
r = -0.78
F1 score
CHGNet
MatterSim-v1.0.0-5M
MACE-MP-0
orb-v2
GRACE-2L-OAM
GRACE-2L-MPtrj
MACE-MPA-0
PET-OAM-XL
NequIP-OAM-L
eSEN-30M-OAM
EquiformerV2
(c)
r = 0.43
κ_{SRME} score
CHGNet
MACE-MP-0
MatterSim-v1.0.0-5M
GRACE-2L-OAM
orb-v2
GRACE-2L-MPtrj
MACE-MPA-0
PET-OAM-XL
NequIP-OAM-L
eSEN-30M-OAM
EquiformerV2